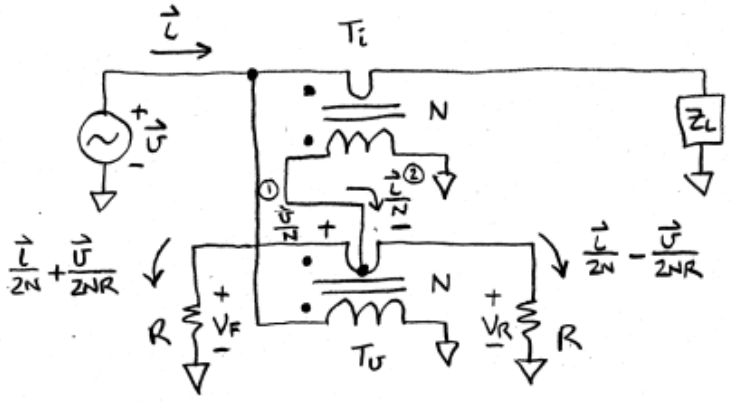
Directional Coupler Analysis

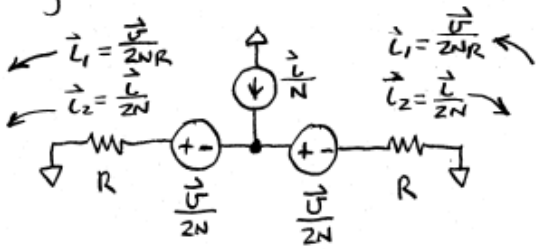
G. Barrere – Exality Corporation



① because of voltage transformer T_V

② because of current transformer T_i

R circuit may be drawn as:



Current through each R is superposition of i-source and v-source currents as shown. Case 1: i-source = 0, case 2: v-sources = 0.

Superposing resulting voltages:

$$\vec{V}_F = \frac{\vec{i}R+\vec{v}}{2NR}$$
$$\vec{V}_R = \frac{\vec{i}R-\vec{v}}{2NR}$$

instantaneous, vector values

Now if current is + leaving source and voltages are referred to ground, the total line current and voltage $\vec{i}$ and $\vec{v}$ may be separated into forward and reverse components:

$$\vec{I} = \vec{I}_F - \vec{I}_R$$

$$\vec{U} = \vec{U}_F + \vec{U}_R$$

So we have

$$\vec{V}_F = \frac{(\vec{I}_F - \vec{I}_R)R + (\vec{U}_F + \vec{U}_R)}{2NR}$$

$$\vec{V}_R = \frac{(\vec{I}_F - \vec{I}_R)R - (\vec{U}_F + \vec{U}_R)}{2NR}$$

and if $R = Z_0 = \frac{\vec{U}_F}{\vec{I}_F} = \frac{\vec{U}_R}{\vec{I}_R}$ = line characteristic impedance, then

$$\vec{V}_F = \frac{(\vec{U}_F - \vec{U}_R) + (\vec{U}_F + \vec{U}_R)}{2NR} = \frac{\vec{U}_F}{NR} = \frac{\vec{I}_F}{N}$$

$$\vec{V}_R = \frac{(\vec{U}_F - \vec{U}_R) - (\vec{U}_F + \vec{U}_R)}{2NR} = \frac{-\vec{U}_R}{NR} = \frac{-\vec{I}_R}{N}$$

These are instantaneous vector quantities, valid for any frequency or waveshape in the forward or reverse directions. They may be detected and scaled appropriately if the waveshape and load are known, to indicate forward and reverse power.